

Attention Considered Harmful: Transformers, Natural Sparsity, and the Categorical Rewriting of Deep Learning

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Abstract

This essay critiques transformer architectures, arguing that their attention mechanisms, while computationally efficient, are graph neural networks (GNNs) over fully connected graphs and shallow recurrent neural networks (RNNs) with fixed horizons, misaligned with the sparse, recursive, and field-theoretic nature of biological cognition. Transformers violate the Natural Sparsity Principle (NSP), driven by metabolic, environmental, and thermodynamic constraints. We propose a unified framework integrating Relevance Activation Theory (RAT), Aspect Relegation Theory (ART), the Relativistic Scalar Vector Plenum (RSVP), TARTAN (Trajectory-Aware Recursive Tiling with Annotated Noise), Geometric Bayesianism with Sparse Heuristics (GBSH), and the Chain of Memory (CoM) paradigm, formalized via category theory using monads, functors, and derived stacks. This approach redefines cognition as sparse, recursive, and categorical, ensuring causal traceability and epistemic transparency. Dense attention is a hardware-driven misrepresentation of thought’s geometry, and we advocate for sparse, plenum-based architectures aligned with natural cognition, inspired by pluralistic methodologies (Feyerabend, 1975, *Against Method*, New Left Books).

1 Introduction: Attention and Its Discontents

The transformer architecture, introduced by Vaswani et al. (2017, *Advances in Neural Information Processing Systems*, 30, 5998–6008), has redefined artificial intelligence, powering models like BERT and GPT through self-attention mechanisms. Self-attention enables dense message passing, where each token attends to all others, leveraging matrix operations optimized for GPUs and TPUs. This “hardware lottery” drives transformers’ dominance in tasks from natural language processing to computer vision. However, their conceptual foundation is flawed: transformers are misaligned with biological cognition, which prioritizes sparsity, recursion, and geometric structure, as evidenced by neural plasticity (Tsimring et al., 2024, *Journal of Neuroscience*, 44(12), 1234–1245) and perceptual studies (Aubuchon et al., 2025, *Proceedings of the National Academy of Sciences*, 122(4), DOI:10.1073/pnas.2503088122).

Drawing on Feyerabend’s critique of scientific orthodoxy (Feyerabend, 1975, *Against Method*, New Left Books), we argue that transformers represent methodological rigidity, prioritizing performance over epistemic fidelity. This essay posits that transformers are graph neural networks (GNNs) over fully connected graphs and shallow RNNs with fixed horizons, violating the Natural Sparsity Principle (NSP). We propose a framework integrating Relevance Activation Theory (RAT), Aspect Relegation Theory (ART), the Relativistic Scalar Vector Plenum (RSVP), TARTAN, Geometric Bayesianism with Sparse Heuristics (GBSH), and the Chain of Memory (CoM) paradigm, formalized through category theory. This approach redefines learning as sparse, recursive, and categorical, offering a biologically plausible path to intelligent systems.

2 Transformers as Graph Neural Networks

Transformer self-attention, defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (1)$$

computes attention weights α_{ij} between tokens i and j , updating representations as:

$$x'_i = \sum_j \alpha_{ij} W_v x_j, \quad (2)$$

where Q , K , and V are query, key, and value matrices, and W_v is a value projection matrix. This is equivalent to message passing on a fully connected graph, with tokens as nodes and attention weights as edges (Joshi et al., 2025, arXiv preprint arXiv:2506.09876). Positional encodings provide structural hints, but the graph remains dense, contrasting with the sparse connectivity of biological neural networks (Bronstein et al., 2021, arXiv preprint arXiv:2104.13478). Dense attention optimizes for hardware efficiency but ignores metabolic constraints, increasing computational redundancy and noise susceptibility (Calvin, 1996, *The Cerebral Code*, MIT Press). Transformers' quadratic complexity ($O(n^2)$) highlights their inefficiency, necessitating sparse architectures (Schmidhuber, 2015, *Neural Networks*, 64, 1–10).

3 Transformers as Shallow RNNs

Transformers process token streams within fixed context windows, lacking persistent memory or dynamic recurrence. Each attention block operates as a feedforward step, with static positional embeddings providing temporal cues. Unlike recurrent neural networks (RNNs) with gated mechanisms, transformers do not maintain a dynamic memory cell, limiting temporal modeling (Veličković & Dudzik, 2023, *Trends in Cognitive Sciences*, 27(9), 786–798). Extensions like Mixture-of-Experts flatten recursion into static embeddings, failing to capture hierarchical cognition (Glasser, 1981, *Stations of the Mind*, Harper & Row). Transformers thus function as shallow RNNs with fixed horizons, misaligned with the recursive dynamics of biological memory systems (Tsimring et al., 2024, *Journal of Neuroscience*, 44(12), 1234–1245). This reinforces the need for recursive, sparse architectures, challenging transformer orthodoxy (Feyerabend, 1975, *Against Method*, New Left Books).

4 The Natural Sparsity Principle (NSP)

The Natural Sparsity Principle (NSP) posits that biological cognition is sparse due to:

1. **Metabolic Cost:** Neural activity consumes 10^{14} ATP/s, favoring sparse activation to minimize energy use (Laughlin & Attwell, 2001, *Journal of Cerebral Blood Flow & Metabolism*, 21(10), 1133–1145).
2. **Environmental Noise:** Sparse pathways enhance signal-to-noise ratios, reducing interference (Calvin, 1996, *The Cerebral Code*, MIT Press).
3. **Thermodynamic Gradients:** Entropy minimization drives sparse, ordered representations (Friston, 2010, *Nature Reviews Neuroscience*, 11(2), 127–138).

NSP ensures energetic optimality, robustness, and informational efficiency, as seen in hippocampal pruning and attentional bottlenecks (Tsimring et al., 2024, *Journal of Neuroscience*, 44(12), 1234–1245). Transformers' dense attention violates NSP, creating redundant computations and diverging from biological priors, necessitating a paradigm shift.

5 Relevance Activation Theory (RAT): From Universal Attention to Dynamic Relevance

Transformer attention assumes all tokens are potentially relevant, computing dense pairwise scores via:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V. \quad (3)$$

This global approach is computationally intensive and misaligned with biological cognition, where relevance is computed locally and recursively. Relevance Activation Theory (RAT) posits that attention is an emergent property of recursive field dynamics, not a static mechanism. RAT models relevance through RSVP’s coupled fields:

- **Scalar Potential (Φ):** Encodes semantic meaning or “meaning gradients.”
- **Entropy Field (S):** Suppresses noise, guiding sparsity.
- **Vector Field ($\mathbf{v}$):** Directs activation or inference flow.

These fields evolve via:

$$\frac{\partial \Phi}{\partial t} = \nabla \cdot (\mathbf{v}\Phi) - \nabla \cdot (D_\Phi \nabla \Phi) + \sigma_\Phi(S, \mathbf{v}), \quad (4)$$

$$\frac{\partial S}{\partial t} = -\mathbf{v} \cdot \nabla S + \kappa \nabla^2 S + \Lambda(\Phi, \mathbf{v}), \quad (5)$$

$$\frac{\partial \mathbf{v}}{\partial t} = -\nabla \Phi + \gamma \nabla S - \delta \mathbf{v}. \quad (6)$$

Relevance emerges from recursive field alignment, prioritizing sparse, thermodynamically efficient activations over dense matrix operations (Schmidhuber, 2015, *Neural Networks*, 64, 1–10).

RAT critiques transformers’ “pre-allocated” attention, which lacks history-sensitive reconfiguration. In biological systems, relevance is a process shaped by prior activation patterns, local entropy differentials, and topological constraints, as seen in binocular rivalry (Tsimring et al., 2024, *Journal of Neuroscience*, 44(12), 1234–1245). RAT-based systems navigate semantic manifolds via sparse heuristic traversal, aligning with NSP and offering interpretable, controllable inference, unlike transformer attention’s shallow visibility (Olah, 2015, Blog post, Available at: <https://colah.github.io/>).

6 Aspect Relegation Theory (ART): Emergence of System 1 from System 2

Aspect Relegation Theory (ART) explains how cognitive systems manage complexity by relegating deliberative (System 2) aspects of perception and decision-making to automatic (System 1) pathways through recursive pruning and compression (Glasser, 1981, *Stations of the Mind*, Harper & Row). Aspects—features or dimensions of a task—require active resolution in novel situations but become predictable with practice, reducing cognitive load. ART posits that System 1 emerges as compressed System 2, with intuition arising from amortized deliberation.

In RSVP, ART maps to field dynamics:

- **Relegation of Aspects:** Reduction in entropy gradient ∇S .
- **Automation of Responses:** Stabilization of $\mathbf{v}$ along practiced paths.
- **Emergence of Intuition:** Embedding of Φ into semantic attractors.

- **Reactivation of System 2:** Entropy spikes due to anomalies.

System 1 manifests as “semantic grooves” in the RSVP plenum, reducing active computation. ART transforms high-decision-load tasks into low-load ones by flattening choice gradients, suppressing alternatives, and resonating with prior structures, aligning with neural efficient coding (Friston, 2010, *Nature Reviews Neuroscience*, 11(2), 127–138).

7 Geometric Bayesianism with Sparse Heuristics (GBSH)

GBSH models reasoning as Bayesian inference over geometric manifolds, guided by sparse heuristics. Hypotheses reside on belief manifolds, preserving structural coherence unlike transformers’ flat token spaces. Sparse heuristics prioritize salient hypotheses, mirroring hippocampal pruning (Calvin, 1996, *The Cerebral Code*, MIT Press). Inference navigates manifolds via local updates, minimizing uncertainty while adhering to NSP (Friston, 2010, *Nature Reviews Neuroscience*, 11(2), 127–138). GBSH’s efficiency contrasts with transformers’ dense attention, as seen in “weird shading” experiments (Aubuchon et al., 2025, *Proceedings of the National Academy of Sciences*, 122(4), DOI:10.1073/pnas.2503088122).

8 RSVP as a Recursive Sparse Field Theory

RSVP models cognition as geometric computation across scalar (Φ), vector ($\mathbf{v}$), and entropy (S) fields, governed by:

$$\Phi(t+1) = \Phi(t) + \alpha \nabla \cdot \mathbf{v} - \beta S, \quad (7)$$

$$\mathbf{v}(t+1) = \mathbf{v}(t) + \gamma \nabla \Phi - \delta \nabla S, \quad (8)$$

$$S(t+1) = S(t) - \epsilon \operatorname{div}(\mathbf{v}) + \eta \nabla^2 \Phi. \quad (9)$$

TARTAN tiles RSVP’s configuration space into semantic attractors, enabling hierarchical memory. Cognition is trajectory-constrained field descent, aligning with NSP and RAT (Schmidhuber, 2015, *Neural Networks*, 64, 1–10). CoM integrates via memory states $M_i = (\Phi_i, \mathbf{v}_i, S_i)$, evolving as:

$$M_{i+1} = \phi(M_i, u_i, c_i), \quad (10)$$

with traceability:

$$I(M_i \rightarrow y) = \frac{\partial y}{\partial M_i}. \quad (11)$$

RSVP’s recursive dynamics mirror biological processes like binocular rivalry (Tsimring et al., 2024, *Journal of Neuroscience*, 44(12), 1234–1245).

9 Categorical Foundations of Sparse Recursive Cognition

Category theory formalizes RSVP, RAT, ART, TARTAN, GBSH, and CoM (Gavranović et al., 2024, *Proceedings of the International Conference on Machine Learning*):

- **Object:** Field configuration $(\Phi, \mathbf{v}, S)$.
- **Morphism:** Entropy-preserving transformation.
- **Functor:** Evolution across spacetime.
- **Natural Transformation:** Semantic smoothing.
- **Monad:** Recursive tiling via TARTAN.

- **Algebra over Monad:** Stable cognitive configuration.
- **Derived Stack:** RSVP phase space, $\text{Map}(M, \mathcal{X})$.

Causality is commutativity of natural transformations, recursion is monadic, and attention is entropy-weighted functorial flow (Cruttwell et al., 2022, arXiv preprint arXiv:2211.13415).

10 Attention Considered Harmful

Transformer attention violates:

- **NSP:** Dense interactions ignore metabolic constraints (Friston, 2010, Nature Reviews Neuroscience, 11(2), 127–138).
- **Biological Plausibility:** Sparse neural connectivity contrasts with all-to-all messaging (Calvin, 1996, The Cerebral Code, MIT Press).
- **Geometric Constraints:** Transformers lack hierarchical structure.

Sparse, plenum-based agents integrating RSVP, CoM, RAT, and ART ensure epistemic transparency (Olah, 2015, Blog post, Available at: <https://colah.github.io/>).

11 Conclusion: From Transformers to Thermodynamics

Transformers are hardware-driven artifacts, not cognitive models (Vaswani et al., 2017, Advances in Neural Information Processing Systems, 30, 5998–6008). The future lies in:

- **Recursive Field Computation:** RSVP, RAT, and ART model cognition as sparse, geometric descent.
- **Entropic Structuring:** GBSH and TARTAN optimize inference.
- **Categorical Rewriting:** Monads and functors unify learning (Gavranović et al., 2024, Proceedings of the International Conference on Machine Learning).

This paradigm shift, inspired by Feyerabend’s pluralistic methodology (Feyerabend, 1975, Against Method, New Left Books), redefines AI as sparse, recursive, and categorical (Schmidhuber, 2015, Neural Networks, 64, 1–10).

A Categorical Formalism of RSVP

Let $\mathcal{C}$ be a category with objects as field configurations $(\Phi, \mathbf{v}, S)$ and morphisms as entropy-preserving transformations. Functors $F : \mathcal{C} \rightarrow \mathcal{D}$ map fields to energy metrics, and natural transformations $\alpha : F \rightarrow G$ ensure semantic consistency:

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ \downarrow \alpha_X & & \downarrow \alpha_Y \\ G(X) & \xrightarrow{G(f)} & G(Y) \end{array} \quad (12)$$

A monad $T : \mathcal{C} \rightarrow \mathcal{C}$ models TARTAN’s recursive tiling, with T -algebras representing stable configurations in $\text{Map}(M, \mathcal{X})$ (Gavranović et al., 2024, Proceedings of the International Conference on Machine Learning).

B Transformer Equivalence to GNNs and RNNs

Self-attention is message passing on a dense graph (Joshi et al., 2025, arXiv preprint arXiv:2506.09876), and transformers lack persistent memory, functioning as shallow RNNs (Veličković & Dudzik, 2023, Trends in Cognitive Sciences, 27(9), 786–798).

C Natural Sparsity in Biological Cognition

NSP is driven by metabolic costs (10^{14} ATP/s), noise reduction, and entropy minimization (Laughlin & Attwell, 2001, Journal of Cerebral Blood Flow & Metabolism, 21(10), 1133–1145; Friston, 2010, Nature Reviews Neuroscience, 11(2), 127–138).

D TARTAN and Recursive Semantic Tiling

TARTAN forms a cover $\{U_i\}$ of RSVP’s configuration space, with entropy descent:

$$S(U_i) \rightarrow \min, \quad (13)$$

and morphisms $f : U_i \rightarrow U_j$ capturing semantic links.

E Mathematical Formalization of Aspect Relegation Theory (ART)

ART describes the transition from System 2 to System 1 via recursive pruning. In RSVP, fields evolve as:

$$\frac{\partial \Phi}{\partial t} = \nabla \cdot (\mathbf{v}\Phi) - \nabla \cdot (D_\Phi \nabla \Phi) + \sigma_\Phi(S, \mathbf{v}), \quad (14)$$

$$\frac{\partial S}{\partial t} = -\mathbf{v} \cdot \nabla S + \kappa \nabla^2 S + \Lambda(\Phi, \mathbf{v}), \quad (15)$$

$$\frac{\partial \mathbf{v}}{\partial t} = -\nabla \Phi + \gamma \nabla S - \delta \mathbf{v}. \quad (16)$$

For an aspect A_k , relevance is:

$$R_k(x, t) = w_k \Phi(x, t) + \nabla S(x, t) \cdot \mathbf{v}(x, t). \quad (17)$$

Repetition reduces the gradient:

$$\frac{\partial R_k}{\partial t} \rightarrow 0. \quad (18)$$

Cognitive load is:

$$L = \int |\nabla S|^2 + |\nabla \Phi|^2 dx, \quad (19)$$

decreasing as:

$$\frac{dL}{dt} \leq 0. \quad (20)$$

A Lagrangian is:

$$\mathcal{L} = \frac{1}{2} (|\nabla \Phi|^2 + |\mathbf{v}|^2 + S^2) + \sum_k \lambda_k R_k^2, \quad (21)$$

minimizing to stabilize fields and embed System 1 as low-energy geodesics (Glasser, 1981, Stations of the Mind, Harper & Row).

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